Multiple Watchman Routes in Staircase Polygons

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Preview

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- An exact algorithm for two watchmen
- Challenges for multiple watchmen
- Dynamic programming approximation and error analysis

Problem Definition

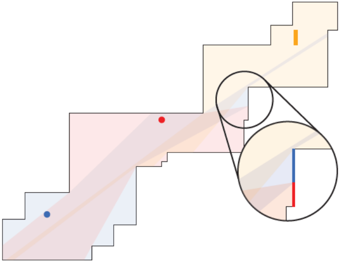
Multiple Watchman Route Problem

Given a polygon P and m watchmen, find a shortest set of m closed routes such that every point in P is visible to at least one route.

Optimization Objectives:

- **Min-Sum:** Minimize total route length $\sum_{i=1}^{m} \|w_i\|$.
- **Min-Max:** Minimize longest route length $\max_{i=1}^{m} \|w_i\|$.

Note. The m-WRP is NP-hard for simple polygons when $m \geq 2$.



Geometric Setting: Staircase Polygons

- **Definition:** Rectilinear polygon with x- and y-monotone chains (”ceiling” & ”floor”).
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- **Vertex Types:**
  - Convex vertex – interior angle $= 90^\circ$
  - Reflex vertex – interior angle $= 270^\circ$
- **Cuts:** directed segments inside P with endpoints on boundary
- **Essential Cuts:** Up to 4 boundary extensions (left, right, top, bottom) that *must* be visited.

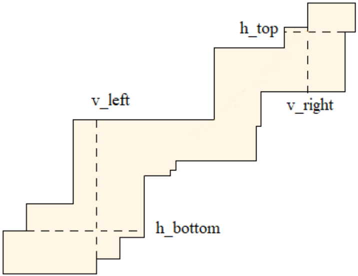

Figure: Staircase polygon with essential cuts.

Review of Prior Work

Single Watchman ($m = 1$):

- Solvable in $O(n^3)$ time for simple polygons (Tan et al.).
- Solvable in $O(n)$ time for simple rectilinear polygons (Chin & Ntafos).

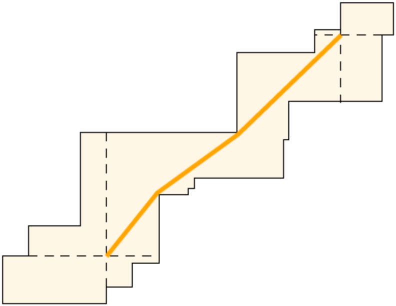

Figure: Optimal solutions for one watchman.

Review of Prior Work

Multiple Watchmen ($m \geq 2$):

- **Hardness:** NP-hard for simple polygons under min-max (Mitchell & Wynters).
- **Approximation:** A 5.969-approximation exists for simple polygons under min-max (Nilsson & Packer).

Motivation

Staircase polygons represent a missing link. Are they restricted enough to allow exact polynomial-time solutions for $m \geq 2$?

Two Watchmen: Structural Properties

Lemma 1 (Route Separation)

There exists an optimal solution (w_1^*, w_2^*) to the 2-WRP such that:

1. w_1^* and w_2^* do not share any common x- or y-coordinates.
2. w_1^* visits the bottom-left cuts $(h_{\text{bot}}, v_{\text{left}})$ and w_2^* visits the top-right cuts $(h_{\text{top}}, v_{\text{right}})$.
3. A diagonal $\overline{rr'}$ between reflex vertices separates w_1^* and w_2^*.

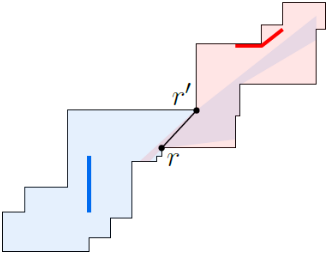

Figure: Optimal solutions for two watchman.

Two Watchmen: Complete Edge Guarding

Unlike general polygons, two watchmen do not need to share monitoring duties for any single boundary segment.

Lemma 2 & 3 (Complete Guarding and Polygon Split)

In an optimal 2-WRP solution, every polygon edge is completely seen by a single watchman. There exists a unique diagonal splitting P into subpolygons P_1 and P_2 such that w_1^* guards P_1 and w_2^* guards P_2.

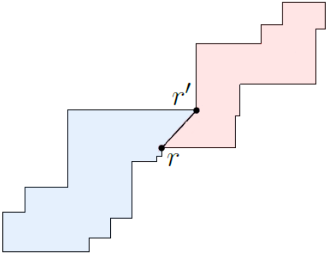

Figure: Decomposition of P into independent subproblems.

Two Watchmen: candidate diagonals

Lemma 4 (candidate diagonals)

All diagonals with negative slope are selected as candidate diagonals. In addition, among the positive-slope diagonals, only those that are immediately adjacent to the negative-slope diagonals in the clockwise order around the floor vertex are selected as candidate diagonals.

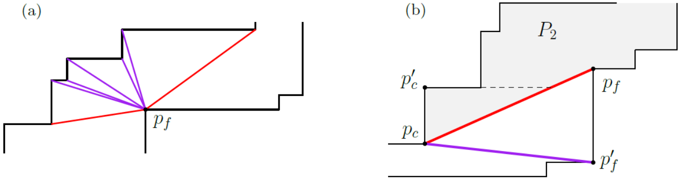

Figure: (a) The candidate diagonals of a reflex vertex pf. (b) A diagonal with positive slope (red) that is not a candidate. An optimal route in P_2 (gray) needs to visit the same essential cut (dashed) as an optimal watchman route in the subpolygon induced by $p_c p_f$ (purple).

Exact Algorithm for Two Watchmen

Algorithm Overview: $O(n^2)$ Time

1. Iterate over $O(n)$ floor vertices to find *candidate diagonals* to the ceiling.
2. For the first diagonal, compute the optimal 1-WRP in each resulting subpolygon in $O(n)$ time.
3. Move clockwise to the next diagonal, maintaining shortest-path-tree data structures.
4. Vertices are added/released in amortized $O(1)$ time per diagonal update.

Reduces the trivial $O(n^3)$ search to an optimal $O(n^2)$ exact algorithm!

Challenges for $m \geq 3$ Watchmen

Why doesn't the diagonal splitting method work for $m \geq 3$?

- **Partial Edge Guarding:** A single boundary edge may be seen partially by different watchmen.
- **Discontiguous Responsibility:** A watchman's visibility region may consist of disconnected parts of the floor and ceiling.
- **No Simple Splits:** Diagonals can no longer guarantee a clean workload partition without losing optimal visibility.

Consequence: We transition from exact algorithms to approximations.

Approximation: Canonical E-Routes ($m \geq 3$)

Definition: Canonical E-Route

A structured route composed of a **left elbow** E, a **right elbow** E', and the shortest paths connecting their respective ceiling and floor endpoints.

- Elbows are formed by intersecting horizontal and vertical extensions of convex vertices.
- **Advantage:** Forces routes to span the full vertical distance between the floor and ceiling, standardizing visibility domains.

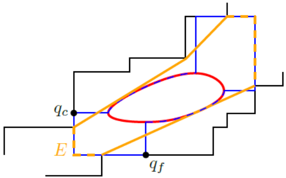

Figure: Transforming an optimal route w (red) into an E-route (orange).

Dynamic Programming & Approximation Bound

We find the optimal set of canonical E-routes using DP:

- Test all $O(n^4)$ pairs of left/right elbows for compatibility ("partnering").
- The recurrence splits the polygon, running in $O(mn^8)$ time.

Theorem (Additive Approximation)

Let W^* be the optimal m-WRP solution, and W be the canonical DP solution. Then:

$$\|W\| \leq \|W^*\| + 4(h_{\mathsf{max}} + v_{\mathsf{max}})$$

where h_{max} and v_{max} are the longest horizontal and vertical line segments in P.

Summary & Future Directions

Key Achievements:

- First structural properties for multi-watchman routes in staircase polygons.
- Exact, polynomial-time $O(n^2)$ algorithm for the 2-WRP.
- Polynomial-time additive approximation for m-WRP ($m \geq 3$).

Open Questions:

- Is the m-WRP strictly NP-hard in staircase polygons for $m \geq 3$?
- Can we find a polynomial-time exact solution for minimum point guarding in this geometric class?

Any Questions?